



Kotebe University of Education
College of Science and Mathematics Education
Department of Mathematics

Mathematics Model Exit Examination-Round Two

Time allowed: 4:00 hours

Choose the correct answer from the given alternatives

1. What is the value of $\lim_{x \rightarrow 0} \frac{3^x - 1}{x}$?

- A. 1
- B. 2
- C. $\ln 2$
- D. Does not exist

2. Consider the function $f(x) = \frac{|x|}{x}$

- A. $\lim_{x \rightarrow 0^+} f(x) = 1$
- B. $\lim_{x \rightarrow 0^-} f(x) = -1$
- C. $\lim_{x \rightarrow 0} f(x)$ does not exist.
- D. All of the above are true about $f(x)$.

3. Which one of the following is true about the function $f(x) = \frac{x - |x|}{x}$?

- A. It is continuous everywhere.
- B. It is not continuous everywhere.
- C. It is continuous at $x = 0$.
- D. It is continuous for all values of x except 0.

4. Let f be a continuous function on the closed interval $[-3, 6]$. If $f(-3) = -1$ and $f(6)$ then the Intermediate Value Theorem guarantees that

- A. $f(0) = 0$
- B. $f'(c) = 4/9$, for at least one c between -3 and 6 .
- C. $-1 \leq f(x) \leq 3$ for all x between -3 and 6 .
- D. $f(c) = 0$ for at least one c between -1 and 3 .

5. If $y = 2^x$, then $\frac{dy}{dx}$ is:

- A. $x(2^{x-1})$
- B. $\frac{2^x}{\log 2}$
- C. $2^x \ln 2$
- D. None of the above

6. The second derivative of $f(x) = \frac{1}{1+x^2}$ is:

- A. $\frac{-2x}{1+x^2}$
- B. $\frac{-2x}{(1+x^2)^2}$
- C. $\frac{-2x^2}{1+x^2}$
- D. $\frac{6x^4 + 4x^2 - 2}{(1+x^2)^4}$

7. A landowner wishes to use 2000 meters of fencing to enclose a rectangular region. Suppose one side of the land lies along a river and does not need fencing. What should be the sides of the region in order to maximize the area?

- A. L=500 m and W= 500 m
- B. L= 1000 m and W=500 m
- C. L= 0 m and W=2000 m
- D. L=1000 m and W=1000 m

8. What is the value of $\int \frac{x}{\sqrt{1+x^2}} dx$?

- A. $\sqrt{1+x^2}$
- B. $\sqrt{1+x^2} + c$
- C. $\sqrt{1+x} + c$
- D. $\arctan(x)$

9. The area of the region bounded by $f(x) = 2\sqrt{x}$, $g(x) = -x$ and line $x = 9$.

- A. 76.5 unit square
- B. 92 unit square
- C. 115 unit square
- D. 112 unit square

10. Let $y = x + \cos x$. Then the value of $\left. \frac{dx}{dy} \right|_{y=0}$ is

- A. 1
- B. 0
- C. $\frac{1}{2}$
- D. -1

11. $\int 4^x dx =$

- A. 4^x
- B. $\frac{4^x}{\ln 4}$
- C. $4^x \ln 4$

- D. $\ln 4$
12. $\lim_{x \rightarrow 0} \frac{\cos x - 1}{x} =$
- A. ∞
B. 1
C. 0
D. -1
13. The series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$ converges to
- A. 0
B. ∞
C. $\frac{1}{2}$
D. 1
14. The interval of convergence for the power series $\sum_{n=1}^{\infty} \frac{x^{2n}}{(-3)^n}$
- A. $(-\sqrt{3}, \sqrt{3})$
B. $(-\infty, \infty)$
C. $(-1, 1)$
D. $(0, 1)$
15. The power series representation of the function $f(x) = \frac{1}{1+x^3}$ is
- A. $\sum_{n=0}^{\infty} (-1)^n (x)^{3n}$
B. $\sum_{n=0}^{\infty} (x)^{3n}$
C. $\sum_{n=0}^{\infty} (-1)^n (x)^n$
D. $\sum_{n=0}^{\infty} (-1)^n (x)^3$
16. The arc length of $y = \ln(\sec(x))$ between $0 \leq x \leq \pi/4$ is
- A. $\ln \sqrt{2}$
B. $\ln(\sqrt{2} + 1)$
C. $\sqrt{2}$
D. 1
17. An approximate value for $\int_0^2 e^{x^2} dx$ using Simpson's rule with $n = 4$ is
- A. 17.3536
B. 20.6445
C. 14.4856
D. 5.6796
18. The Taylor series representation for the function $f(x) = e^x$ about $x = 0$ is
- A. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$
B. $\sum_{n=0}^{\infty} n! x^n$
C. $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$

D. $\sum_{n=0}^{\infty} \frac{(-3)^n x^n}{n!}$

19. Which of the following is True about the function $(z) = \frac{e^z + (iz)^2}{z^2}$?

- A. It is continuous and differentiable.
- B. It is continuous but not differentiable.
- C. It is Neither continuous nor differentiable.
- D. It is differentiable but not continuous.

20. Which of the following is **Not** an analytic function?

- A. xy
- B. $x + y$
- C. $2xy + i(x^2 + y^2)$
- D. $2xy$

21. What is the value of the integral of $g(z)$ around the circle $|z - i| = 2$ in the positive sense when $g(z) = \frac{1}{z^2 + 4}$?

- A. $\frac{\pi}{16}$
- B. π
- C. $\frac{\pi}{4}$
- D. $\frac{\pi}{2}$

22. Which of the following curves are correct on which the function $f(z) = (x - y)^2 + 2i(x + y)$

- A. $x - y = 1$
- B. $x + y = 1$
- C. $x^2 + y^2 = 1$
- D. $x^2 - y^2 = 1$

23. Which one of the following functions has a pole?

- A. $f(z) = \frac{\sin z}{z}$
- B. $f(z) = \frac{1}{(z-3i)^5}$
- C. $f(z) = e^{1/z}$
- D. None of the above

24. Let $f(z) = \frac{1}{z^2 + z}$. What is the residue at $z = 0$?

- A. 3
- B. 4

- C. $\frac{1}{2}$
D. 1

25. The power series representation of the function $g(z) = \frac{1}{z^2}$ is :-

- A. $\sum_{n=0}^{\infty} (-1)^n (z-1)^n, \quad (|z-1| < 1)$
B. $\sum_{n=1}^{\infty} (-1)^n n (z-1)^{n-1}, \quad (|z-1| < 1)$
C. $\sum_{n=0}^{\infty} (-1)^n (n+1)(z-1)^n, \quad (|z-1| < 1)$
D. $\sum_{n=1}^{\infty} (-1)^n (n-1)(z-1)^n, \quad (|z-1| < 1)$

26. What is the integral of $f(z) = \frac{z+1}{z^2-2z}$ around the circle $|z|=3$ in the positive sense using Cauchy's residue theorem?

- A. $2\pi i$
B. $-2\pi i$
C. $\frac{2}{3} \pi i$
D. πi

27. Which one of the following is the Laurent series representation of the function $f(z) = z^2 \sin \frac{1}{z^2}$ in the domain $0 < |z| < \infty$?

- A. $\sum_{n=1}^{\infty} \frac{(-1)^{-n}}{(2n+1)!} \cdot \frac{1}{z^{2n}}$
B. $1 + \sum_{n=1}^{\infty} \frac{(-1)^{-n}}{(2n+1)!} \cdot \frac{1}{z^{4n}}$
C. $1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{(2n)!}$
D. $\sum_{n=1}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot \frac{1}{z^{2n}}$

28. Which one of the following is the **range** of the real valued function $f(x, y) = \sqrt{1 - x^2 - y^2}$?

- A. $\{(x, y): x^2 + y^2 \leq 1\}$
B. $\{z: 0 \leq z \leq 1\}$
C. $\{(x, y): x^2 + y^2 \geq 1\}$
D. $\{z: -1 \leq z \leq 1\}$

29. The equation of the tangent line to the surface $z = xe^y$ at $(1, 0, 1)$ is:

- A. $z = x + y$
B. $z = -x - y$
C. $z = x - y$
D. $z = -x + y$

30. Let $f(x, y) = x^2 + 4y^2$. Then the directional derivative of f in the direction of $\left(\frac{3}{5}, \frac{4}{5}\right)$ is:

- A. 4
B. $-\frac{8}{5}$
C. -8
D. $\frac{44}{5}$

31. The equation of the line passing through the point $(-2, 5, -3)$ and perpendicular to the plane

$$2x - 3y + 4z + 7 = 0 \text{ is:}$$

- A. $(2, -3, 4) + t(-2, 5, -3)$
B. $(2, -3, 4) + t(-2, 5, -3)$

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- A. 4
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- C. -8
- D. $\frac{44}{5}$

31. The equation of the line passing through the point $(-2, 5, -3)$ and perpendicular to the plane $2x - 3y + 4z + 7 = 0$ is:

- A. $(2, -3, 4) + t(-2, 5, -3)$
- B. $(2, -3, 4) + t(-2, 5, -3)$
- C. $(-2, 5, -3) + t(2, -3, 4)$
- D. $(-4, 3, -7) + t(2, -3, 4)$

32. Let R be the rectangular region bounded by the lines $x = -1$, $x = 2$, $y = 0$, and $y = 2$.

Then $\iint_R x^2 y dA$ is equal to:

- A. 4
- B. 8
- C. 5
- D. 6

33. $\lim_{t \rightarrow 0} (2 \cos t \mathbf{i} - \frac{\sin t}{t} \mathbf{j} + t^2 \mathbf{k})$ is equal to:

- A. $2\mathbf{i} - \mathbf{j}$
- B. $-\mathbf{j}$
- C. $-\mathbf{j} + 2\mathbf{k}$
- D. $2\mathbf{i}$

34. Which one of the following planes in $\mathbb{R}^3$ is perpendicular to yz -plane?

- A. $x + y - 1 = 0$
- B. $2x - 3z + 2 = 0$
- C. $-3y + 2z - 4 = 0$
- D. D) $6x - z + 1 = 0$

35. Let $(x, y) = x \int_0^y e^{t^2} dt$. Then $\frac{\partial f}{\partial y}$ is equal to:

- A. xe^{y^2}
- B. e^{x^2}
- C. e^y
- D. ye^x

36. If $f(x, y) = 5xy + \sin(y - 2x)$ then, $\nabla f(1, 2)$ is:

- A. $(5, 3)$

- B. (-2, 10)
- C. (1, 2)
- D. D) (8, 6)

37. Let $f(x, y) = e^{xy}$. Then $\frac{\partial^2 f}{\partial y \partial x}$ is equal to:

- A. xe^{xy}
- B. y^2e^{xy}
- C. $e^{xy}(xy + 1)$
- D. x^2e^{xy}

38. If we use three-digit rounding arithmetic to compute $\frac{\frac{13}{14} - \frac{6}{7}}{2e - 5.4}$, which of the following statements is

NOT true?

- A. The result is 1.80
- B. The relative error is 0.0788
- C. The absolute error is 0.154
- D. None of these

39. What is the rate of convergence of the secant method of solving non-linear equations?

- A. 1
- B. 2
- C. 1.618
- D. 2.618

40. Which of the following is an iterative formula to find $\sqrt{2}$?

- A. $x_{n+1} = x_n + \frac{2}{x_n}$
- B. $x_{n+1} = \frac{1}{2} \left(x_n + \frac{2}{x_n} \right)$
- C. $x_{n+1} = 2 \left(x_n + \frac{2}{x_n} \right)$
- D. $x_{n+1} = \frac{1}{2} \left(x_n + \frac{1}{x_n} \right)$

41. Which of the following statements is true about iterative methods of solving systems of equations?

- A. They are more effective than direct methods.
- B. They always converge to the solution.

- C. They are self-correcting methods.
- D. All of these

42. Consider the following table.

x	0	1	2	3
$y = f(x)$	4	3	?	7

If $(E - 1)^3 f(x) = 0$, then the missing term is equal to

- A. -4
- B. 4
- C. 7
- D. 12

43. The linear interpolating polynomial for the data points (1, 3) and (4, 6) is

C. They are self-correcting methods.

D. All of these

42. Consider the following table.

x	0	1	2	3
$y = f(x)$	4	3	?	7

If $(E - 1)^3 f(x) = 0$, then the missing term is equal to

A. -4

B. 4

C. 7

D. 12

43. The linear interpolating polynomial for the data points (1, 3) and (4, 6) is

A. $p(x) = x + 2$

B. $p(x) = 3x + 2$

C. $p(x) = x + 6$

D. $p(x) = \frac{1}{3}x + \frac{2}{3}$

44. Using trapezoidal rule with two subintervals, what is the value of $\int_0^1 \frac{dx}{1+x}$ correct to 2-decimal places?

A. 0.71

B. 0.70

C. 0.69

D. 0.6

45. Which one of the following is the solution of the system of linear equations?

$$x_1 + 2x_2 + 3x_3 = 6$$

$$3x_1 + 2x_2 + 9x_3 = 14$$

$$2x_1 + 4x_2 + x_3 = 7$$

A. [2, -1, 2]

B. [1, 1, 1]

C. [-2, 2, 2]

D. [3, 3, -1]

46. A tailor has 80 sq.m of cotton material and 120 sq.m of woolen material. A suit requires 1sq.m of cotton and 3 sq.m of woolen material and a dress requires 2 sq.m of cotton and 2 sq.m of woolen material. A suit sells for birr 7000 and a dress sells for birr 5000. Then the tailor desires to maximize his profit. Then which of the following is the correct mathematical formulation of the problem(l.p.p)?

A. Maximize, $z = 7000x_1 + 5000x_2$

Subject to

- $x_1 + 2x_2 \leq 80$
 $3x_1 + 2x_2 \leq 120 \quad x_1, x_2 \geq 0$
- B. Maximize , $z = 7000x_1 + 5000x_2$
 Subject to
 $x_1 + 2x_2 \geq 80$
 $3x_1 + 2x_2 \leq 120 \quad x_1, x_2 \geq 0$
- C. Maximize , $z = 7000x_1 + 5000x_2$
 Subject to
 $x_1 + 2x_2 \leq 80$
 $3x_1 + 2x_2 \geq 120 \quad x_1, x_2 \geq 0$
- D. Maximize , $z = 7000x_1 + 5000x_2$
 Subject to
 $x_1 + 2x_2 \geq 80$
 $3x_1 + 2x_2 \leq 120 \quad x_1 \in \Re, x_2 \in \Re$

47. Which one of the following is **not** convex set?

- A. Hyperplane
 B. $S = \{x: |x| \leq 1\}$
 C. Intersection of two convex sets.
 D. In $\mathbb{R}^2$, $S = \{(x, y); y^2 \geq 4x\}$

Direction: Answer questions 48, 49 and 50 based on the following linear programming problem and its optimal table.

Given Maximize, $z = x_1 - x_2 + 3x_3$

(Primal problem) Subjected to

$$\begin{aligned}
 x_1 + x_2 + x_3 &\leq 10 \\
 2x_1 - x_3 &\leq 2 \\
 2x_1 - 2x_2 + 3x_3 &\leq 0 \quad x_1, x_2, x_3 \geq 0
 \end{aligned}$$

So that the optimality of the problem remaining undisturbed. The optimal simplex table given by

Basis	c_B	b	a_1	a_2	a_3	a_4	a_5	a_6
a_2	-1	6	$\frac{1}{5}$	1	0	$\frac{3}{5}$	0	$-\frac{1}{5}$
a_5	0	6	$\frac{14}{5}$	0	0	$\frac{2}{5}$	1	$\frac{1}{5}$
a_3	3	4	$\frac{4}{5}$	0	1	$\frac{2}{5}$	0	$\frac{1}{5}$
$z_j - c_j$		6	$\frac{6}{5}$	0	0	$\frac{3}{5}$	0	$\frac{4}{5}$

48. What is the optimal value of the primal problem?

- A. 24
 B. 12
 C. 18

D. 6

49. The optimal solution of the dual problem is

- A. $[u_1 = \frac{3}{5}, u_2 = 0, u_3 = \frac{4}{5}]$
- B. $[u_1 = \frac{1}{5}, u_2 = 0, u_3 = \frac{1}{5}]$
- C. $[u_1 = 0, u_2 = \frac{2}{5}, u_3 = \frac{4}{5}]$
- D. $[u_1 = \frac{3}{5}, u_2 = \frac{1}{5}, u_3 = \frac{4}{5}]$

50. Which of the following is not true?

A. $B^{-1} = \begin{bmatrix} \frac{3}{5} & 0 & -\frac{1}{5} \\ \frac{2}{5} & 1 & \frac{1}{5} \\ \frac{2}{5} & 0 & \frac{1}{5} \end{bmatrix}$

- B. The range of b_3 changes given by the interval $[-20 \leq b_3 \leq 30]$
- C. For $b_3 = -20$, the B.F.S corresponding to the basis B is $B^{-1}[10, 2, -20]$ is $\{10, 2, 0\}$
- D. For $b_3 = 30$, the B.F.S corresponding to the basis B is $B^{-1}[10, 2, 30]$ is $\{10, 12, 6\}$

51. Which one of the following is not true about Duality theory?

- A. The dual of the dual problem is primal problem.
- B. The coefficient matrix of the dual is the transpose of the coefficient matrix of the primal
- C. Minimization of the objective function of the dual transform to maximization of the objective function of the primal
- D. If the k^{th} constraints of a primal be an equation, the k^{th} dual variable will be non-negative.

52. Which one of the following is true about both model and reality?

- A. Both mimic relevant features of the circumstances.
- B. Both mimic the circumstances by using the language of mathematics.
- C. Model mimic reality by using the language of mathematics.
- D. Reality mimic model by using the language of mathematics.

53. Consider the census of a certain city population is given below

Years (time)	Population size in million (N)
1993	1
2003	2.5
2013	3.8
2023	5

Which of the following mathematical equation expresses the above argument?

A. $\frac{dN}{dt} = -\alpha N$

B. $\frac{dN}{dt} = -\alpha N^2$

C. $\frac{dN}{dt} = \alpha N$

D. $\frac{dN}{dt} = \alpha N^2$

54. Which one of the following model is dimensionally incorrect?

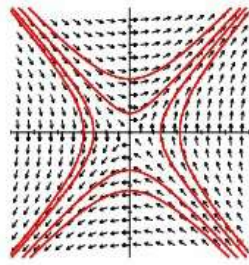
A. $S = \frac{1}{2}gt^2$

B. $F = \frac{Gm_1m_2}{r}$

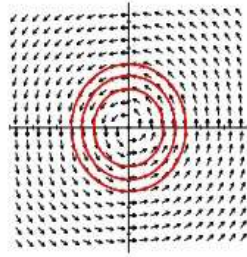
C. $F = \frac{Gm_1m_2}{r^2}$

D. $F = ma$

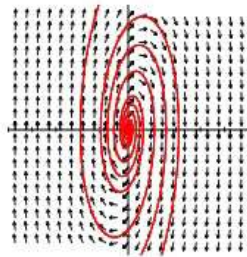
55. Which one of the following graph represents a saddle point?



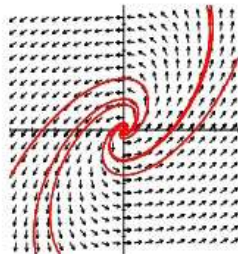
A.



B.



C.



D.

56. Suppose that the Lotka – Volterra system

$$\frac{dP}{dt} = -bPR + aP$$

$$\frac{dR}{dt} = -cR + dPR$$

with initial conditions $P(t_0) = P_0$ and $R(t_0) = R_0$ where $P(t)$ is the number of prey; $R(t)$ is the number of predator; and the parameters $a, b, c, d > 0$. Then what is the possible equilibria for the system?

- A. $(0, 0)$ and $\left(-\frac{c}{d}, \frac{a}{b}\right)$
- B. $(0, 0)$ and $\left(-\frac{c}{d}, -\frac{a}{b}\right)$
- C. $(0, 0)$ and $\left(\frac{a}{b}, \frac{c}{d}\right)$
- D. $(0, 0)$ and $\left(\frac{c}{d}, \frac{a}{b}\right)$.

57. What is the stability for the linear system $\begin{cases} \frac{dx}{dt} = x - y, \\ \frac{dy}{dt} = x + y - 2 \end{cases}$?

- A. Stable spiral
- B. Center
- C. Unstable spiral
- D. Stable node

58. For which values of m is the function $y = x^m$ a solution of the differential equation

$$x^2 y'' - 5xy' + 8Y = 0?$$

- A. 2 and 3
- B. 3 and 4
- C. 2 and 4
- D. D. -2 and -4

59. Let $xy' + 3y = x^2$ be a first-order linear differential equation. Then what is the integrating factor of the equation?

- A. $\mu(x) = e^{4x}$
- B. $\mu(x) = x^3$
- C. $\mu(x) = 4 \ln x$
- D. $\mu(x) = e^{\frac{4}{x}}$

60. Let $y_1 = e^{2t}$ and $y_2 = te^{2t}$. Which of the following statements is true about $W(y_1, y_2)$?

- A. $W(y_1, y_2) = e^{4t}$, therefore y_1 and y_2 are linearly independent.
- B. $W(y_1, y_2) = e^{4t}$, therefore y_1 and y_2 are linearly dependent.
- C. $W(y_1, y_2) = 0$, therefore y_1 and y_2 are linearly independent.
- D. $W(y_1, y_2) = 0$, therefore y_1 and y_2 are linearly dependent.

61. Consider the second-order nonhomogeneous differential equation: -

$$y'' - 4y' + 3Y = e^t + t^2$$

What is the complementary solution?

- A. A. $y_c = c_1 e^t + c_2 t^2$
- B. $y_c = c_1 e^t + c_2 e^{3t}$
- C. $y_c = c_1 e^{-t} + c_2 e^{-3t}$
- D. $y_c = c_1 e^t + c_2 e^{-3t}$

62. Let $f(t) = e^{at}$, its Laplace transform for $s > a$ is given by: -

- A. $\frac{a}{s^2} + (s - a)$

- B. $\frac{1}{s-a}$
 C. $\frac{\sqrt{\pi}}{2(s-a)}$
 D. does not exist
63. The solution of the differential equation $x\sqrt{1+y^2}dx + y\sqrt{1+x^2}dy = 0$ is
 A. $\sin^{-1}x + \sin^{-1}y = c$
 B. $\tan^{-1}x + \tan^{-1}y = c$
 C. $\sqrt{1+x^2} + \sqrt{1+y^2} = c$
 D. $\tan^{-1}x + \sqrt{1+y^2} = c$
64. Which one of the following describes the Fourier series of a function described by
 $f(x) = x + \pi, -\pi < x < \pi$ and $f(x + 2\pi) = f(x)$
 A. $\frac{\pi}{2} + \sum_{n=1}^{\infty} -\frac{2}{n} \sin(n\pi)(\cos nx)$
 B. $\pi + \sum_{n=1}^{\infty} -\frac{2}{n} \cos(n\pi)(\sin nx)$
 C. $\frac{\pi}{2} + \sum_{n=1}^{\infty} -\frac{2}{n} \cos(n\pi)(\sin nx)$
 D. $\pi + \sum_{n=1}^{\infty} -\frac{2}{n} \sin(n\pi)(\cos nx)$
65. What is the value of $\int_0^{\infty} e^{-x} \cos 2x dx$
 A. 0
 B. 1
 C. $\frac{1}{5}$
 D. $\frac{2}{5}$
66. If the integral $\int_{-a}^a f(x)dx = 2 \int_0^a f(x)dx$, then which one of the following is true about $f(x)$.
 A. f is an odd function
 B. f is neither even nor odd
 C. f is an even function
 D. f is a polynomial function
67. The second order partial differential equation $u_{xx} + xu_{yy} = 0$ is:-
 A. Elliptic for $x > 0$
 B. Hyperbolic for $x > 0$
 C. Elliptic for $x < 0$
 D. Hyperbolic for $x = 0$
68. What is the order of the equation $xy^3(\frac{\partial y}{\partial x})^2 + yx^2 + \frac{\partial y}{\partial x} = 0$
 A. Third order
 B. Second order
 C. Zero order
 D. First order
69. What is the general solution of the linear partial differential equation $yzp + zxq = xy$?
 A. $\phi(x^2 - y^2 - z^2) = 0$
 B. $\phi(x^2 - y^2, y^2 - z^2) = 0$
 C. $\phi(x^2 - y^2, y^2 - x^2) = 0$
 D. $\phi(x^2 - z^2, z^2 - x^2) = 0$

- D. $\phi(x^2 - z^2, z^2 - x^2) = 0$
70. What is the Fourier series transform of e^{-ax} ?
 A. $\frac{a}{s^2+a^2}$
 B. $\frac{s}{s^2+a^2}$
 C. $\frac{a}{s^2-a^2}$
 D. $\frac{s}{s^2-a^2}$
71. The solution of the partial differential equation $u_{xy} = -u_x$ is:-
 A. $u(x, y) = f(x)e^{-y} + g(y)$
 B. $u(x, y) = f(y)e^{-y} + g(x)$
 C. $u(x, y) = f(x)e^y + g(y)$
 D. $u(x, y) = f(y)e^y + g(x)$

D. $\varphi(x^2 - z^2, z^2 - x^2) = 0$

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A. $\frac{a}{s^2 + a^2}$

B. $\frac{s}{s^2 + a^2}$

C. $\frac{a}{s^2 - a^2}$

D. $\frac{s}{s^2 - a^2}$

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C. $u(x, y) = f(x)e^y + g(y)$

D. $u(x, y) = f(y)e^y + g(x)$

72. What is the general solution of the partial differential equation $x^2 z_x + y^2 z_y = -(x + y)z$?

A. $\varphi\left(\frac{1}{x} - y, (x - y)z\right) = 0$

B. $\varphi\left(\frac{1}{x} + \frac{1}{y}, xy + z\right) = 0$

C. $\varphi\left(\frac{1}{x} - \frac{1}{y}, (x - y)z\right) = 0$

D. $\varphi\left(x - y, \left(\frac{1}{x} - y\right)z\right) = 0$

73. The Fourier integral of Fourier transforms method is used for _____ functions.

A. Odd

B. Even

C. Periodic

D. Non-periodic

74. What does the face value of a bond represent?

A. The current market value of the bond.

B. The value of the bond when it matures.

C. The interest rate of the bond.

D. The price paid to purchase the bond initially.

75. The yield to maturity (YTM) of a bond is:

A. The interest rate stated on the face of the bond.

B. The annual interest payment divided by the face value of the bond.

C. The total return anticipated on a bond if it is held until it matures.

D. The market price of the bond.

76. What does a higher IRR indicate about an investment?

A. A lower risk associated with the investment.

B. A higher risk associated with the investment.

C. Higher profitability and efficiency of the investment.

D. Lower profitability and efficiency of the investment.

77. What is the formula for the expectation (mean) of a continuous random variable X with probability density function f(x)?

A. $E(X) = \int x * f(x) dx$

- B. $E(X) = \sum x * f(x)$ over all possible values of x
- C. $E(X) = \int f(x) dx$
- D. $E(X) = \sum f(x)$ over all possible values of x
78. For a continuous random variable X and a function $g(X)$, how is the expectation of $g(X)$ calculated?
- A. $E(g(X)) = g(E(X))$
- B. $E(g(X)) = g(\int x * f(x) dx)$
- C. $E(g(X)) = \int g(x) * f(x) dx$ over all possible values of x
- D. $E(g(X)) = g(\int f(x) dx)$ over all possible values of x
79. In the context of arbitrage, what is the riskless profit that arises from market inefficiencies known as?
- A. Market speculation.
- B. Market equilibrium.
- C. Market arbitrage.
- D. Market volatility.
80. Which one of the following is **NOT** true about any group structure?
- A. Identity element is unique.
- B. Cancellation laws hold.
- C. For any two elements a and b , $(ab)^{-1} = b^{-1}a^{-1}$.
- D. Finite abelian groups are cyclic groups.
81. Which one of the following is True?
- A. Every subgroup of a group should contain the identity element.
- B. Every subring of a ring with unity should contain the unity.
- C. The order of a subgroup divides the order of the group.
- D. Every group of order 6 is abelian group.
82. Let $\mathbb{Q}^*$ be the set of all nonzero rational numbers. Define Δ on $\mathbb{Q}^*$ by $a\Delta b = \frac{ab}{5}$. Which one of the following is **NOT** true?
- A. $(\mathbb{Q}^*, \Delta)$ is an infinite group
- B. The inverse of 2 in $(\mathbb{Q}^*, \Delta)$ is $\frac{1}{2}$.
- C. The identity element in $(\mathbb{Q}^*, \Delta)$ is 5.
- D. $(\mathbb{Q}^*, \Delta)$ is an abelian group.
83. Which one of the following is **NOT** necessarily an abelian group?
- A. A group of order 4
- B. A group of order 5
- C. A group of order 9
- D. A group of order 15
84. Which one of the following is **NOT** true?
- A. Every integral domain is a field.
- B. Every field is a unique factorization domain.
- C. Every field has only two ideals.
- D. The only maximal ideal of a field is $\{0\}$.

85. Which one of the following is **NOT** an integral domain?

- A. The ring $(\mathbb{Z}, +, \cdot)$
- B. The ring $(\mathbb{Z}_2, \oplus_2, \odot_2)$
- C. The ring $(\mathbb{Z}_4, \oplus_4, \odot_4)$
- D. The ring $(\mathbb{Z}_5, \oplus_5, \odot_5)$

86. Which one of the following is true about the ring $(\mathbb{Z}, +, \cdot)$?

- A. All of its ideals are of infinite order.
- B. All of its ideals are principal ideals.
- C. The only unit in $\mathbb{Z}$ is 1.
- D. It is a division ring.

87. Which one of the following is the correct values of the scalars a, b and c satisfying $X = aU + bV + cW$ for $X = (-1, -2, -2)$, $U = (0, 1, 4)$, $V = (-1, 1, 2)$ and $W = (3, 1, 2)$ in $\mathbb{R}^3$?

- A. $a = 2, b = -1$ and $c = -1$
- B. $a = -1, b = 2$, and $c = 1$
- C. $a = 1, b = -2$ and $c = -1$
- D. $a = 2, b = -1$ and $c = 1$

88. Which one of the following is true?

- A. The vectors $(\sqrt{2} \ \sqrt{2} \ \sqrt{3})^T$ and $(2 \ 2 \ \sqrt{6})^T$ are parallel vectors in $\mathbb{R}^3$.
- B. The vectors $(\frac{1}{2} \ \frac{-1}{2} \ \frac{\sqrt{2}}{2})^T$ and $(\frac{1}{\sqrt{2}} \ \frac{-1}{\sqrt{2}} \ 0)^T$ are both unit vectors in $\mathbb{R}^3$.
- C. The distance between the vectors $(a \ -\sqrt{2}a \ 0)^T$ and $(-2a \ 0 \ \sqrt{5}a)^T$ is $4|a|$, $a \in \mathbb{R}$.
- D. The basis of some three dimensional vector spaces has two linearly independent vectors.

89. If $u = (2 \ 0 \ -6)^T$ and $v = (a + b \ b - c \ 3c)^T \in \mathbb{R}^3$ are orthogonal, then which one of the following is true?

- A) $a = 2b$
- B) $c = \frac{a}{2}$
- C) $b = 2$
- D) $b = -c$

90. If $A = -i + 3j + k$ and $B = 2i + 4j + 3k$, then which one of the following is correct?

- A. $Proj_B^A = \left(\frac{26}{29}, \frac{52}{29}, \frac{39}{29}\right)$
- B. $Proj_A^B = \left(-\frac{13}{121}, \frac{39}{121}, \frac{13}{121}\right)$
- C. $comp_A^B = \frac{13}{11}$

D. $\text{comp}_A^B = \frac{13}{29}$

91. Let $v = \sqrt{2}i + 3j - \sqrt{10}k$ be a vector in $\mathbb{R}^3$. If this vector makes an angle of α , β and γ with the positive x , y and z axes respectively. Then the **directional cosines** of v , which can be computed as follows:

A. $\cos \alpha = \frac{1}{2\sqrt{2}}$

B. $\cos \beta = 1$

C. $\cos \gamma = \frac{5}{2}$

D. $v = 4(\cos \gamma, \cos \beta, \cos \alpha)$

92. Which one of the following is true about the matrix $A = \begin{bmatrix} 3 & 1 & -5 \\ 0 & 0 & 0 \\ 0 & 4 & 0 \end{bmatrix}$?

A. A has three distinct eigenvalues.

B. The eigenvector associated with the eigenvalue $\lambda = 3$ is $v = (-1 \ 0 \ 0)^T$.

C. It has one eigenvector and two generalized eigenvectors.

D. A has only two linearly independent vectors associated to all its eigenvalues.

93. Which one of the following is **false** for an m by n matrix A in reduced row echelon form?

A. All zero rows of A , if any, are at the bottom.

B. The first non-zero entry in any row appears in a column to the right of the first non-zero entry in any preceding row.

C. The first non-zero entry in any non-zero row is one.

D. Every column of A is a unit vector.

94. Which one of the following is false?

A. In the ring $(2\mathbb{Z}, +, \bullet)$, 2 divides 10.

B. In a commutative ring R with unity, if every non-zero element of R is a unit, then these non-zero element of R are the divisor of its every nonzero element.

C. If R is a commutative ring with unity and $a, b, c \in R$ such that a divides b and a divides c , then a divides $bx + cy$ for any $x, y \in R$.

D. If a, b , and c are integers such that $\gcd(b, a) = c$ and b divides ac , then b necessarily divides c .

95. Let $x = (237)_{\text{eight}}$ and $y = (345)_{\text{six}}$ then which one of the following is false?

A. $x + y = 296$

B. $x + y = (1212)_{\text{six}}$

C. $x - y = (26)_{\text{eight}}$

D. $x + y = (258)_{\text{nine}}$

96. Which one of the following is not an integral solution of $18x + 7y = 302$

- A. $x = 2$ and $y = 38$
- B. $x = 9$ and $y = 20$
- C. $x = 2$ and $y = 18$
- D. $x = 18$ and $y = 2$

97. Which one of the following is correct about?

- A. The triple $(1, 4, 7)$ is the primitive solution of the Diophantine equation $x^2 + 3y^2 = z^2$.
- B. All the solution of the Diophantine equation $x^2 + 2y^2 = z^2$ is triple $(k, 2k, 3k)$ for any $k \in \mathbb{Z}$.
- C. The triple $(5, 20, 35)$ is a Pythagorean triple for the Diophantine equation $x^2 + 3y^2 = z^2$.
- D. The primitive solution for the Diophantine equation $x^2 + 2y^2 = z^2$ is the triple $(3, 6, 9)$.

98. Which one of the following is the remainder when $2^{82} + 1$ is divided by 7?

- A. 4
- B. 5
- C. 0
- D. 3

99. Consider the congruence $9x^{17} + 3x^{10} + 3x^9 - 2x^2 + 2 \equiv 0 \pmod{15}$. Then which one of the following is **false**?

- A. $x^8 \equiv x \pmod{15}$
- B. The congruence given is equivalent to the congruence $x^2 + 12x + 2 \equiv 0 \pmod{15}$.
- C. The complete solution of the given congruence are $\equiv 1 \pmod{15}$, $x \equiv 2 \pmod{15}$, $x \equiv 7 \pmod{15}$ and $x \equiv 11 \pmod{15}$.
- D. $9x^{17} + 3x^{10} - 2x^2 \equiv (x^2 + 9x) \pmod{15}$.

100. Which one of the following is false?

- A. Solving the congruence $x^3 - x^2 + 7x + 1 \equiv 0 \pmod{200}$ is equivalent to solving the system $\begin{cases} x^3 - x^2 + 7x + 1 \equiv 0 \pmod{8} \\ x^3 - x^2 + 7x + 1 \equiv 0 \pmod{25} \end{cases}$
- B. The congruence $x^3 + 2x^2 + 4x + 3 \equiv 0 \pmod{7}$ is equivalent to the congruence $(x - 1)(x^2 + 3x + 2) \equiv 0 \pmod{7}$.

- C. The congruence $x^3 + 2x^2 + 4x + 3 \equiv 0 \pmod{7}$ has the same solution as the congruence $x^3 + 2x^2 + 6x + 5 \equiv 0 \pmod{7}$.
- D. The system of congruence $x \equiv b_i \pmod{m_i}$ for pairwise relative prime numbers m_i 's $\forall i = 1: n$, has a unique solution modulo m , where $m = \gcd(m_1, m_2, \dots, m_n)$.

Answer Key

- | | | |
|---------------|---------------|----------------|
| 1. Answer: C | 18. Answer: A | 32. Answer: D |
| 2. Answer: D | 19. Answer: A | 33. Answer: A |
| 3. Answer: D | 20. Answer: C | 34. Answer: C |
| 4. Answer: C | 21. Answer: D | 35. Answer: A |
| 5. Answer: C | 22. Answer: A | 36. Answer: D |
| 6. Answer: D | 23. Answer: B | 37. Answer: C |
| 7. Answer: B | 24. Answer: D | 38. Answer: D |
| 8. Answer: B | 25. Answer: C | 39. Answer: C |
| 9. Answer: A | 26. Answer: A | 40. Answer: B |
| 10. Answer: A | 27. Answer: B | 41. Answer: C |
| 11. Answer: B | 28. Answer: B | 42. Answer: B |
| 12. Answer: C | 29. Answer: A | 43. Answer: A |
| 13. Answer: D | 30. Answer: A | 44. Answer: A |
| 14. Answer: A | 31. Answer: C | |
| 15. Answer: A | 65. Answer: C | 86. Answer: B |
| 16. Answer: B | 66. Answer: C | 87. Answer: C |
| 17. Answer: A | 67. Answer: A | 88. Answer: D |
| 45. Answer: B | 68. Answer: D | 89. Answer: A |
| 46. Answer: A | 69. Answer: B | 90. Answer: A |
| 47. Answer: D | 70. Answer: B | 91. Answer: A |
| 48. Answer: D | 71. Answer: A | 92. Answer: B |
| 49. Answer: A | 72. Answer: C | 93. Answer: D |
| 50. Answer: D | 73. Answer: D | 94. Answer: A |
| 51. Answer: D | 74. Answer: B | 95. Answer: D |
| 52. Answer: C | 75. Answer: C | 96. Answer: C |
| 53. Answer: C | 76. Answer: C | 97. Answer: D |
| 54. Answer: B | 77. Answer: A | 98. Answer: B |
| 55. Answer: A | 78. Answer: C | 99. Answer: A |
| 56. Answer: D | 79. Answer: C | 100. Answer: D |
| 57. Answer: C | 80. Answer: D | |
| 58. Answer: C | 81. Answer: A | |
| 59. Answer: B | 82. Answer: B | |
| 60. Answer: A | 83. Answer: D | |
| 61. Answer: B | 84. Answer: A | |
| 62. Answer: B | 85. Answer: C | |
| 63. Answer: C | | |
| 64. Answer: B | | |